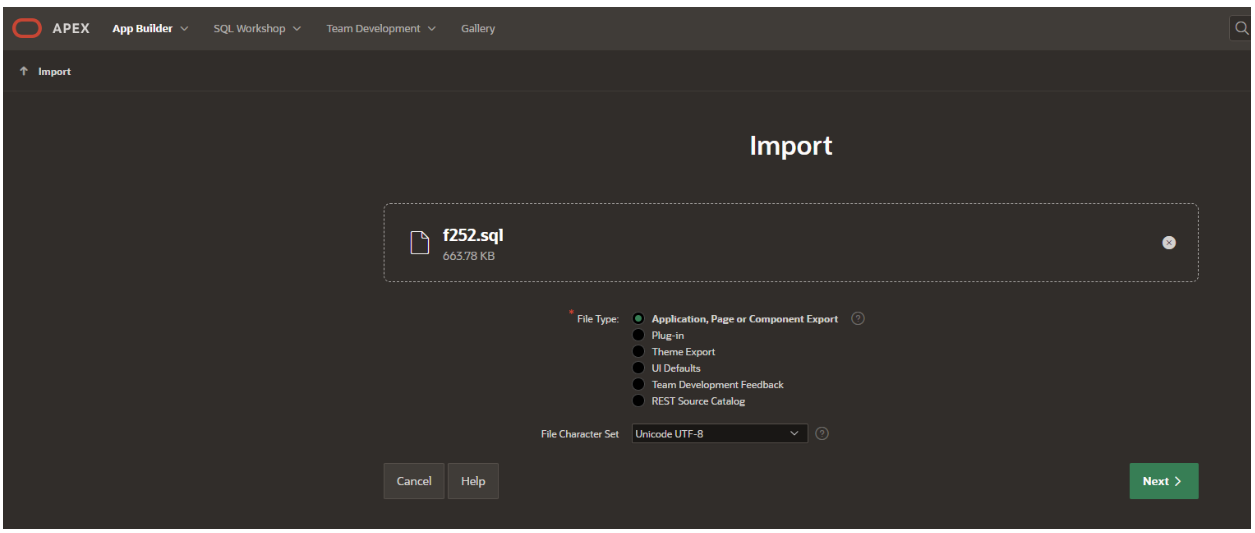
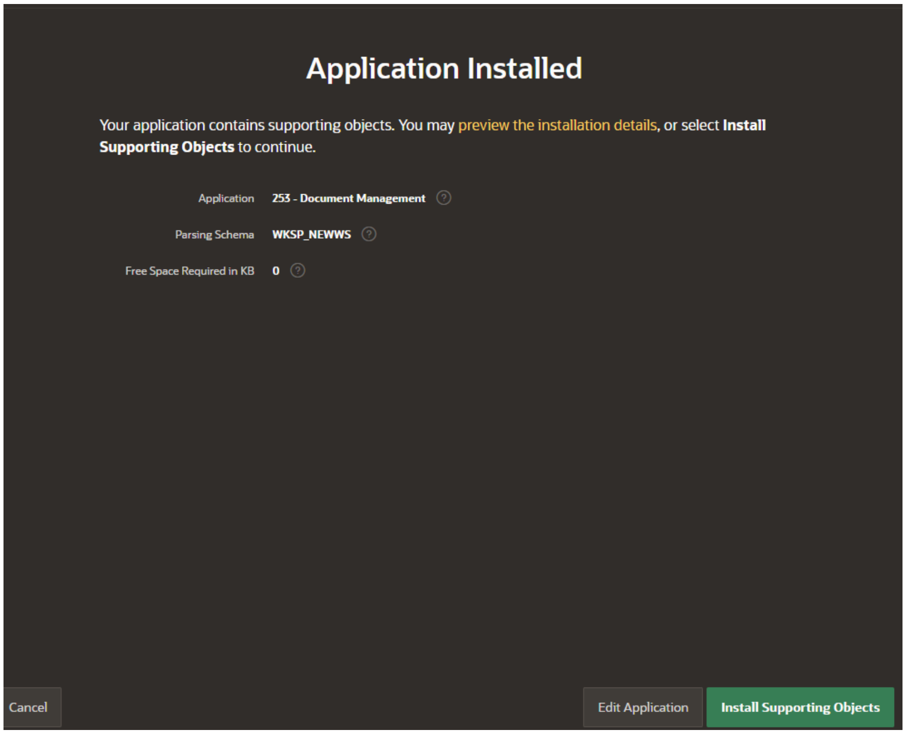
Document Management App – Installation Guide

Step 1. Application Installation

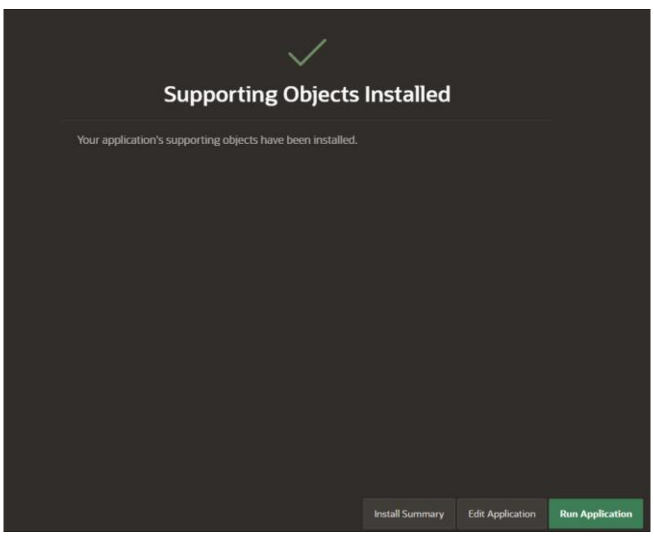
1. Create a workspace or use an existing one.
2. Login to APEX workspace.
3. Go to App Builder -> Import. Drag & Drop the app file and click on **Next** and follow the prompts to upload the app.



4. APEX will prompt to **Install Supporting Objects**. You can review the Install Summary. Once the objects are installed, a success message will appear.



5. Click **Run Application** to launch the app.



Step 2 Additional Configuration

1. This application is using Vector Search, a feature of Oracle Database 26ai. You can import and ONNX model into the Oracle Database and generate embeddings. You need to login as an administrator and grant the necessary privileges to the parsing schema: create mining model and execute on dbms_vector and dbms_cloud packages.

```
grant create mining model to YourSchemaName;
grant execute on dbms_vector to YourSchemaName;
grant execute on dbms_cloud to YourSchemaName;
```

2. Next step is to load the ONNX model into the database. Refer to the documentation to download the model (all-MiniLM-L12-v2), unzip and load it. Make sure to give “**DOC_MODEL**” as model name.

Example of loading the model stored in an Object Storage bucket, using a pre-authenticated request:

```
DECLARE
  L_PAR_URL      VARCHAR2(1000);
  L_RESPONSE_BLOB BLOB;
  BEGIN
    L_PAR_URL := 'INSERT_HERE_THE_LINK_TO_MODEL'
    ;
    L_RESPONSE_BLOB := APEX_WEB_SERVICE.MAKE_REST_REQUEST_B(
      P_URL        => L_PAR_URL,
      P_HTTP_METHOD => 'GET'
    );
    DBMS_VECTOR.LOAD_ONNX_MODEL(
      MODEL_NAME => 'DOC_MODEL',
      MODEL_DATA => L_RESPONSE_BLOB,
      METADATA   =>
           JSON(
             '{
           "function" : "embedding",
           "embeddingOutput" : "embedding" ,
           "input":{"input": ["DATA"]}
         }'
           )
    );

  END;
```

Example of loading the model stored in an Object Storage bucket, using credentials:

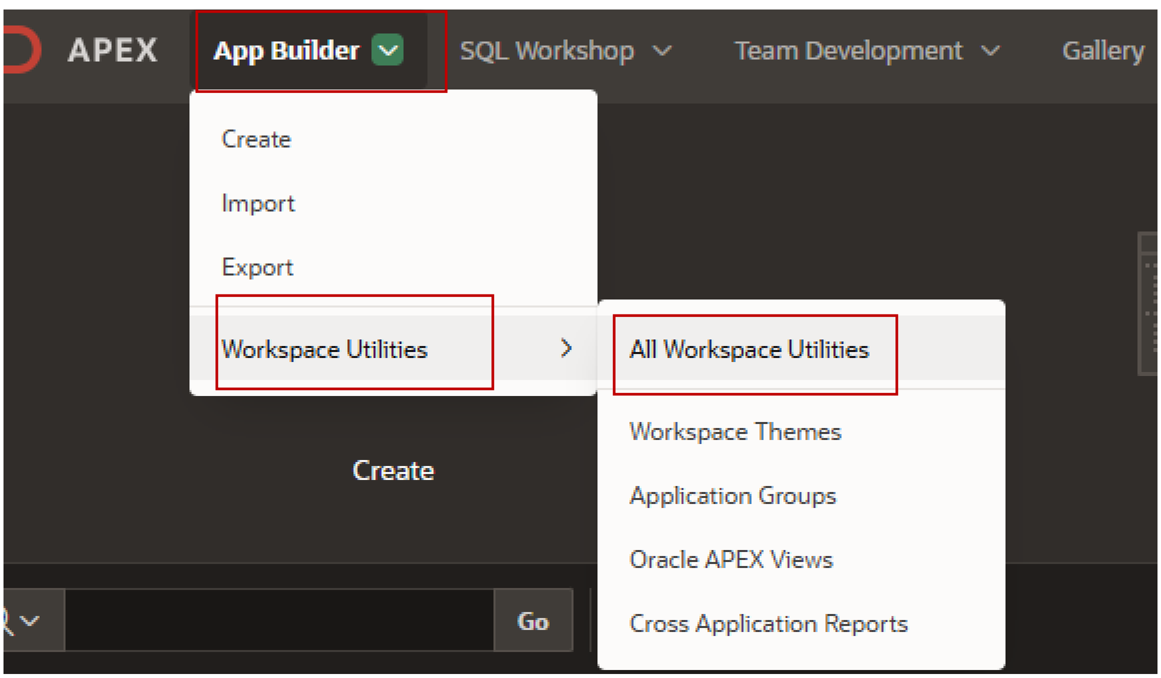
- *Create the credentials*

```
BEGIN
 dbms_cloud.create_credential (
  credential_name => 'onnx_obj_store_cred',
  username       => '<Your username>',
  password       => '<AUTH Token>'
 );
END;
```

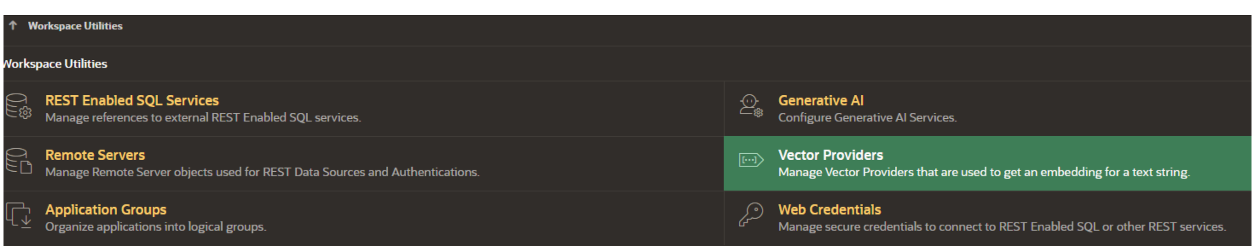
- *Load the model*

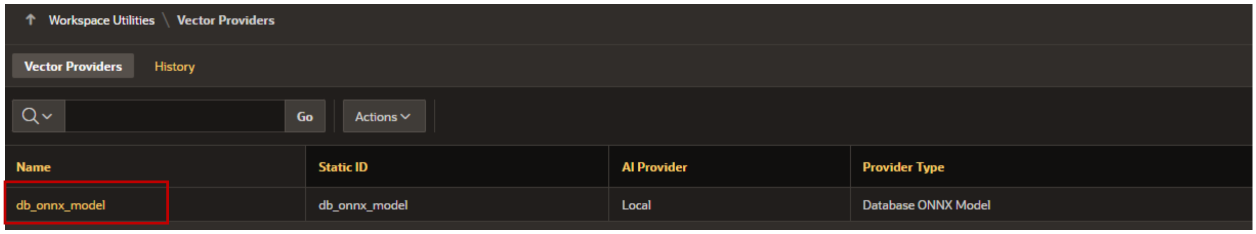
```
BEGIN
DBMS_VECTOR.LOAD_ONNX_MODEL(
        model_name => 'DOC_MODEL',
        model_data => dbms_cloud.get_object (
         credential_name => 'obj_store_cred', -- replace with your credential name
         object_uri      => '<Enter Your Object Storage URI>'), -- blob
        metadata => JSON('{
           "function" : "embedding",
           "embeddingOutput" : "embedding" ,
           "input":{"input": ["DATA"]}
        }')
      );
END;
```

3. Modify the vector provider that was imported to point to your parsing schema. Learn more about vector providers from [this](#) blog. Click on **App Builder** -> **Workspace Utilities** -> **All Workspace Utilities.**



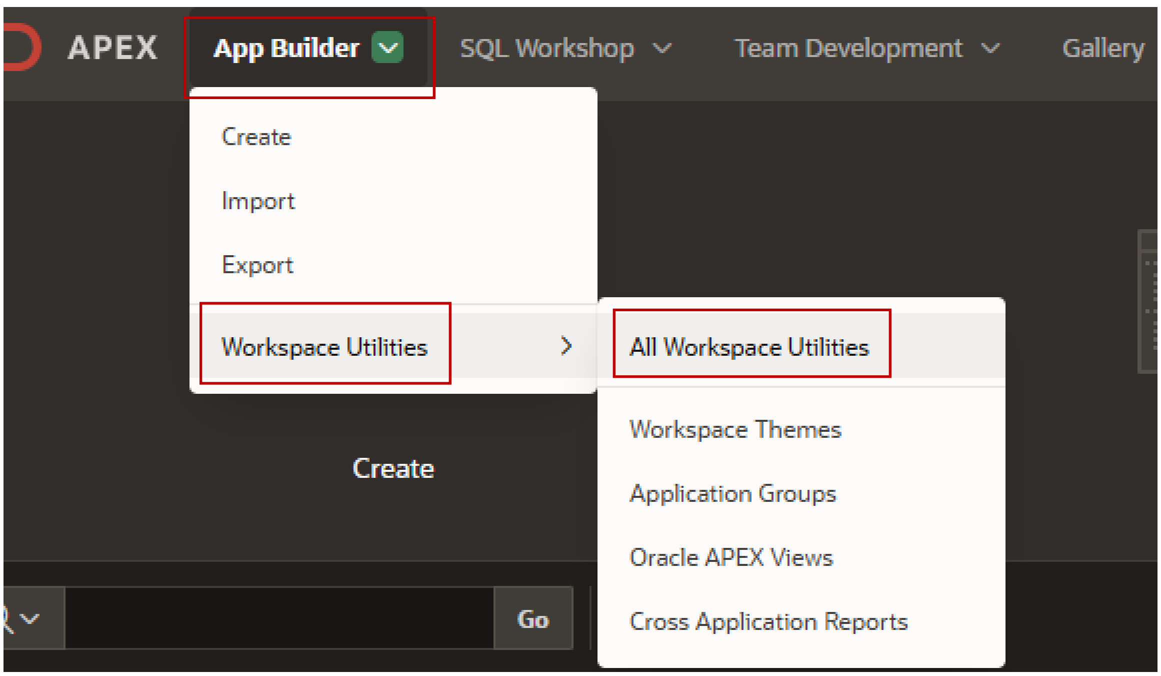
Click on **Vector Providers** and select **db_onnx_model**.



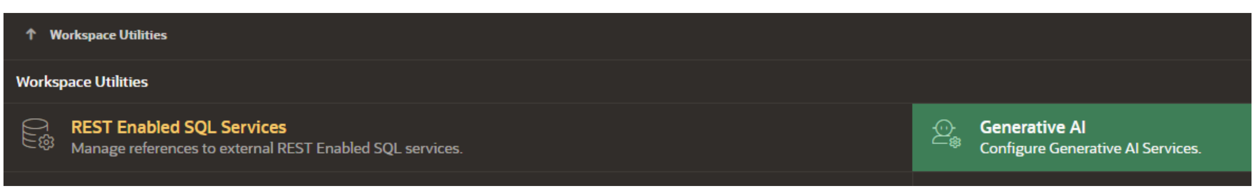


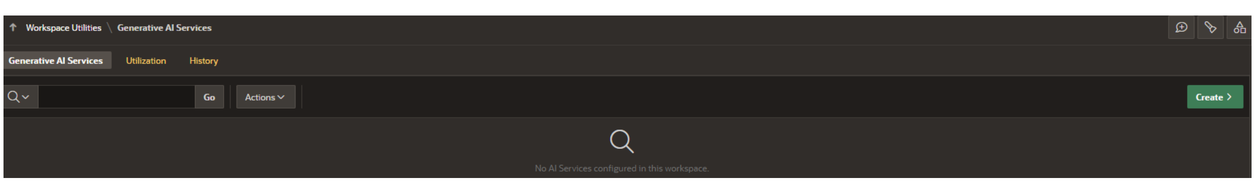
Modify the **ONNX Model Owner** to point to your schema and select the **ONNX Model Name** imported earlier. Click on **Apply Changes**.

4. The application leverages APEX AI capabilities for end users. It is using the two AI dynamic actions: Generate Text with AI and Show AI Assistant. To use these capabilities, you need to configure Generative AI Services at the workspace level. You have three options: OCI Generative AI, Open AI and Cohere. Click on **App Builder** -> **Workspace Utilities** -> **All Workspace Utilities.**

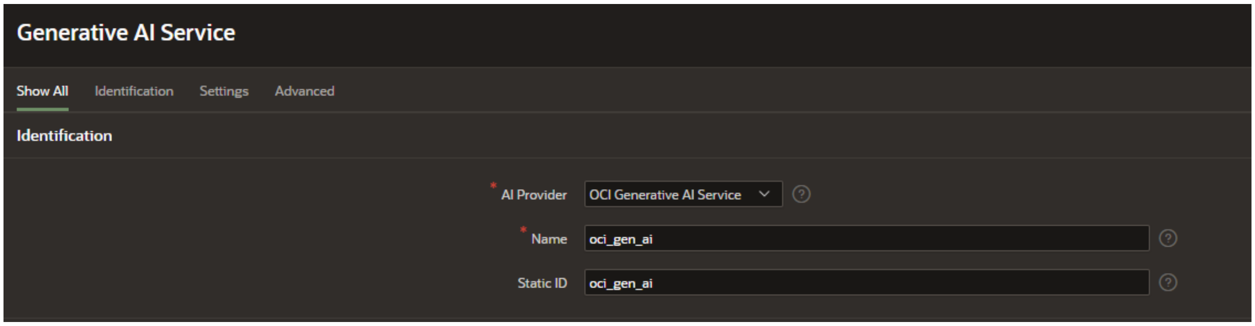


Click on **Generative AI** and then on **Create** button.

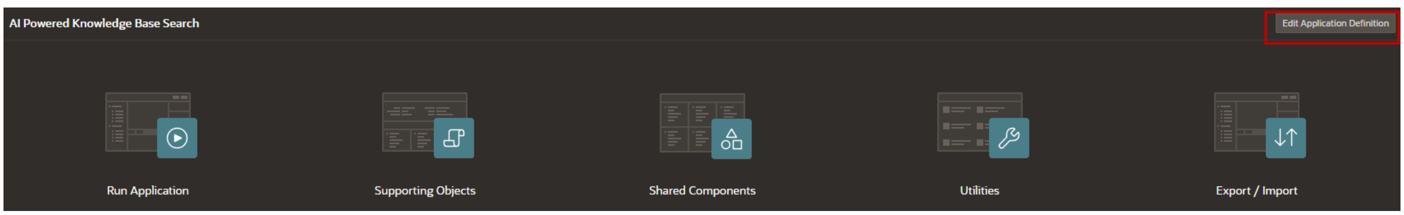




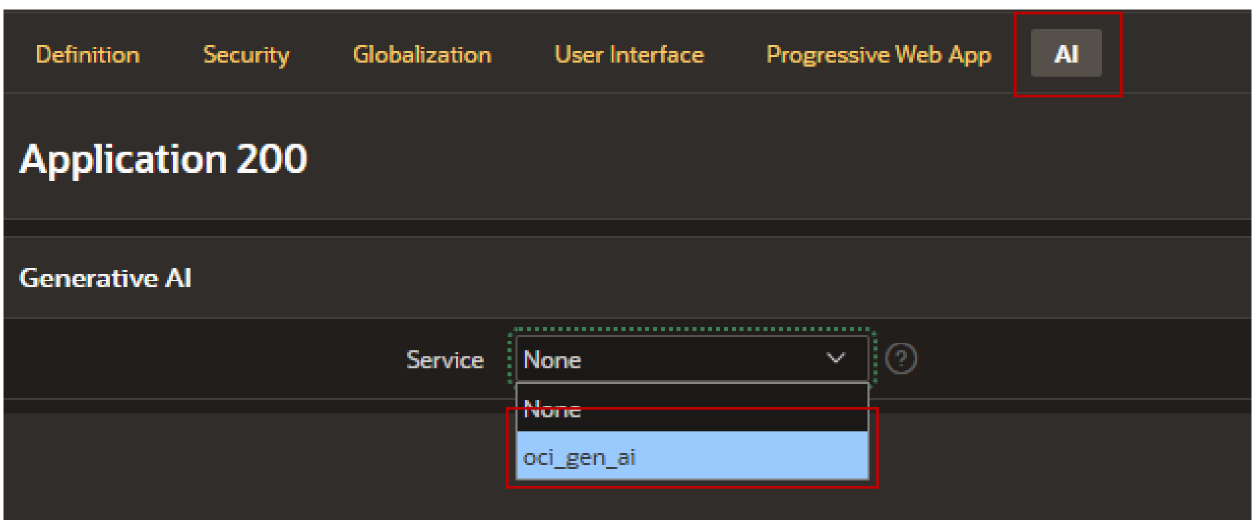
Select your AI Provider and enter the required details. Refer to the documentation for additional information. Type “oci_gen_ai” for name and static id.



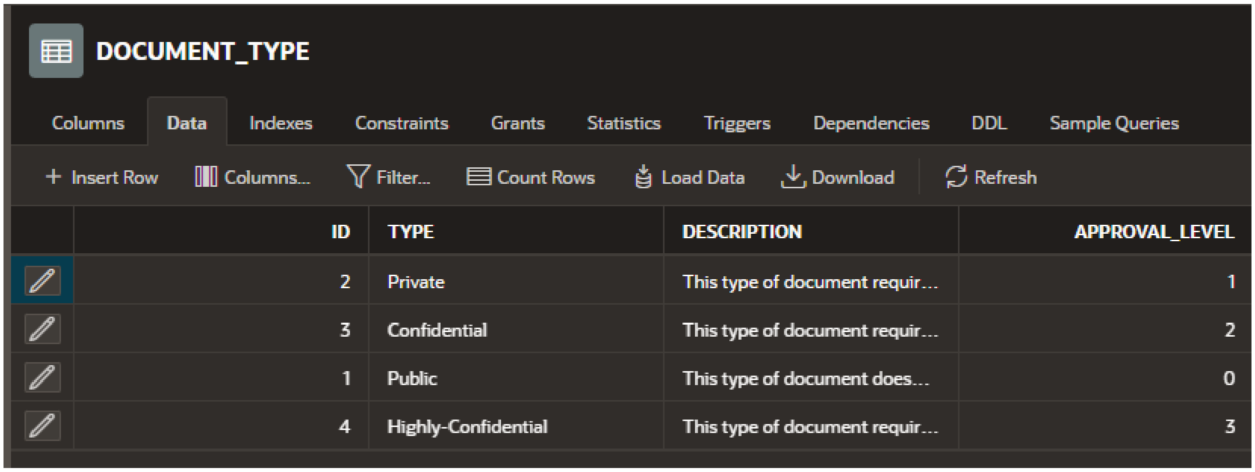
Edit the application definition to have this Generative AI Service as default service. Open the application in App Builder. Click on **Edit Application Definition**.



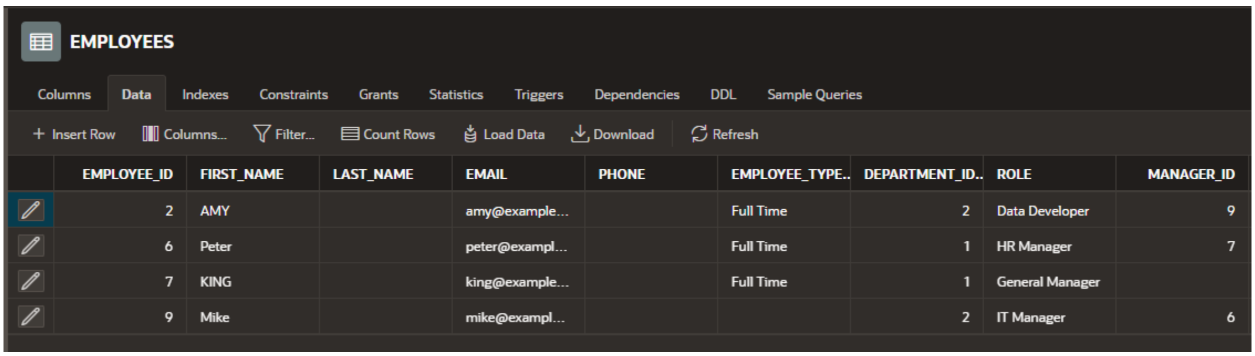
Click on **AI** tab and select the oci_gen_ai service. Then, click on **Apply Changes**.



5. Depending on the document type, different approval levels may be required. The DOCUMENT_TYPE table defines each document type and its corresponding number of approval levels, while the EMPLOYEE table stores the team or organizational hierarchy. For example, if a document is classified as Confidential, the required approval level is 2. This means the document must be approved by both the employee’s direct manager and the next-level manager.

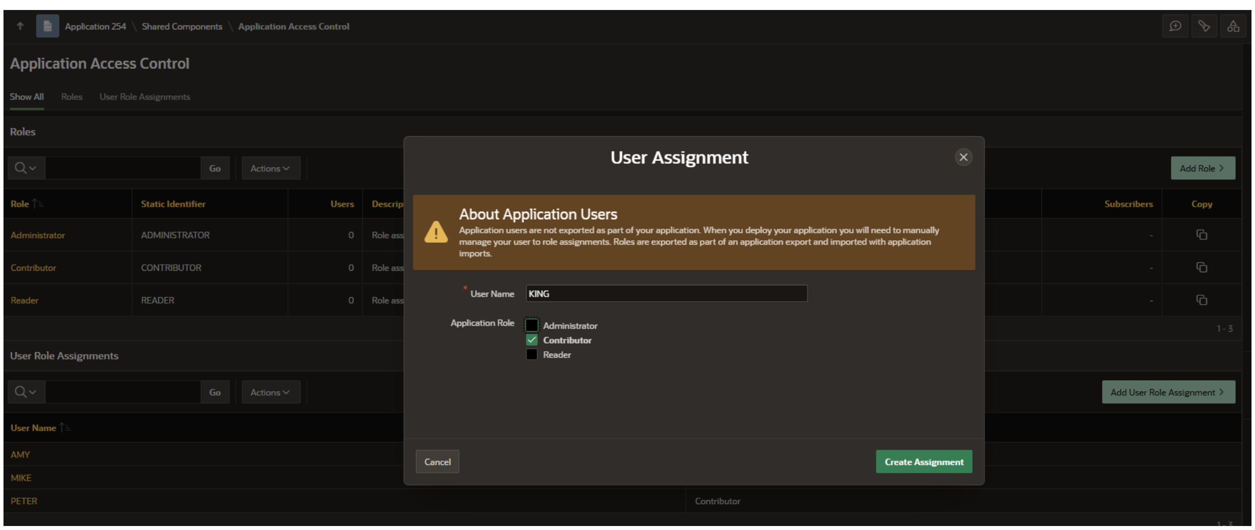


ID	TYPE	DESCRIPTION	APPROVAL_LEVEL
2	Private	This type of document requir...	1
3	Confidential	This type of document requir...	2
1	Public	This type of document does...	0
4	Highly-Confidential	This type of document requir...	3



EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	EMPLOYEE_TYPE..	DEPARTMENT_ID..	ROLE	MANAGER_ID
2	AMY		amy@example...		Full Time	2	Data Developer	9
6	Peter		peter@exampl...		Full Time	1	HR Manager	7
7	KING		king@example...		Full Time	1	General Manager	
9	Mike		mike@exampl...			2	IT Manager	6

You can create the users **AMY**, **MIKE**, **PETER**, and **KING**, then navigate to **App XXX → Shared Components → Application Access Control.** From there, add each user and assign them the appropriate role, such as Contributor or Administrator.



That's it! You are ready to run and explore the application. Upload PDF documents and ask questions!